भारतीय सूचना प्रौद्योगिकी संस्थान गुवाहाटी
INDIAN INSTITUTE OF INFORMATION TECHNOLOGY GUWAHATI

CS351: IT Workshop II (Cloud Computing), End Semester Examination, Monsoon 2024

Note: Maximum time: 3 Hours. Total Marks: 60

- Answer all of the following questions. Answers must be brief and to the point.
- Make reasonable assumptions wherever necessary and state them.

1. (a) [2 points] Map Reduce can help in achieving scalability. Justify how.
   (b) [4 points] We have a server log having different HTTP status codes (e.g., 200, 404). Write pseudocodes for mapper and reducer to find the count of different HTTP status codes.
2. (a) [3 points] Write about three main features of Hadoop Distributed File System (HDFS).
   (b) [2 points] Write the differences between HDFS and HBase.
   (c) [2 points] What are the different schedulers available in Hadoop?
3. (a) [2 points] How Apache Hive performs integration with Hadoop Ecosystem?
   (b) [3 points] Write a user-defined function (UDF) in Apache Pig to find the length of a text field (say name) for each record of a dataset.
4. (a) [3 points] Explain the architecture of Apache Spark with a diagram.
   (b) [3 points] What are the functions of collect() and map() in Apache Spark?
5. (a) [3 points] How does Single Sign-On (SSO) based authentication work?
   (b) [3 points] Asymmetric encryption can help in securing data in cloud. Justify.
6. (a) [3 points] You have a dataset having many features of the objects. Write map and reduce functions (pseudo-codes) to perform grouping of these objects.
   (b) [3 points] Explain with an example how Apache Spark helps in classifying big data.
7. (a) [4 points] How can we distribute multimedia files globally using cloud computing? How can cloud perform media transcoding here?
   (b) [2 points] Write about the advantages of two main streaming protocols present in multimedia cloud.
8. (a) [2 points] Write about the main features of Docker Containers.
   (b) [4 points] You have an web application written in Python. Write a Dockerfile and create a docker image from the Dockerfile. Write the commands needed.
9. (a) [3 points] How can two docker containers communicate with each other? Explain.
   (b) [3 points] What are the pods and deployments in Kubernetes? How do they work together?
10. (a) [3 points] How micro-services are deployed in cloud computing? Explain.
    (b) [3 points] How do we benchmark a cloud application? Explain the steps.